**K J SOMAIYA INSTITUTE OF TECHNOLOGY**
**Department of Information Technology**

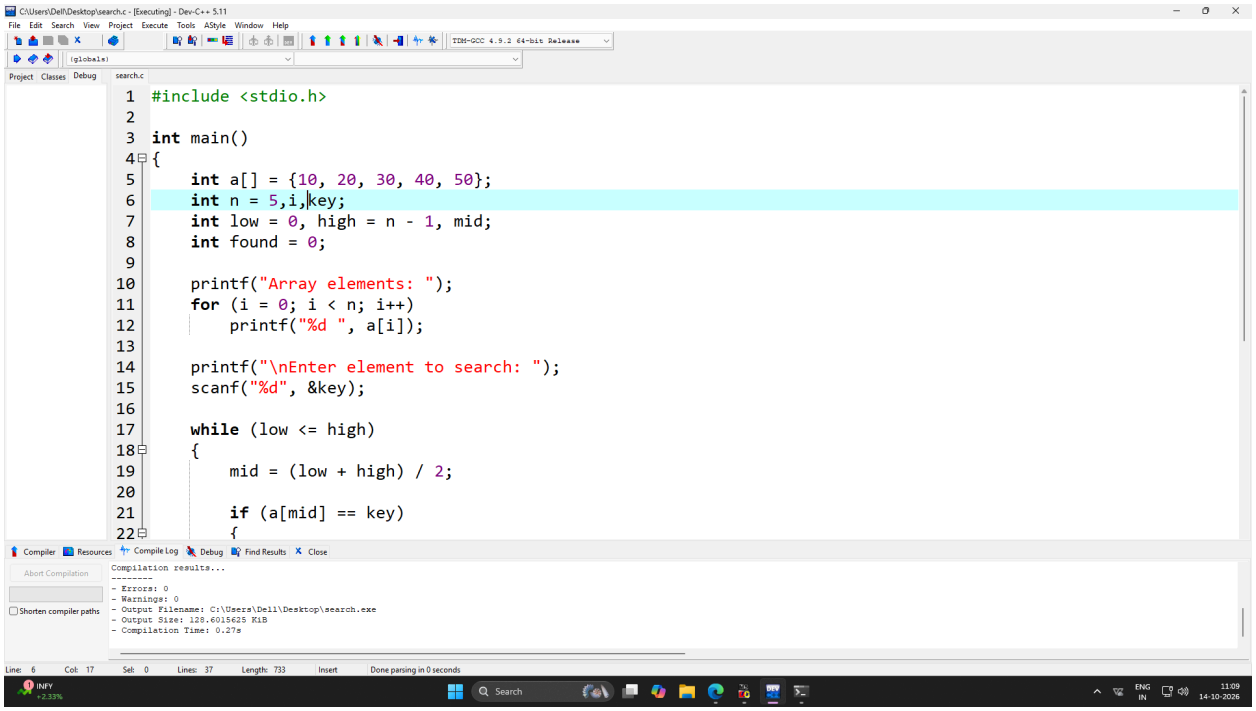
**Academic Year: 2026-27**
**Class: S.Y. B.Tech.**
**Semester: III**

**Subject: DSA LAB**
**Course Code: ITL302**

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**Division: IT-A**
**Batch: SA2**
**Roll Number: 27**

EXPERIMENT NO:09

Program:



```c
#include <stdio.h>

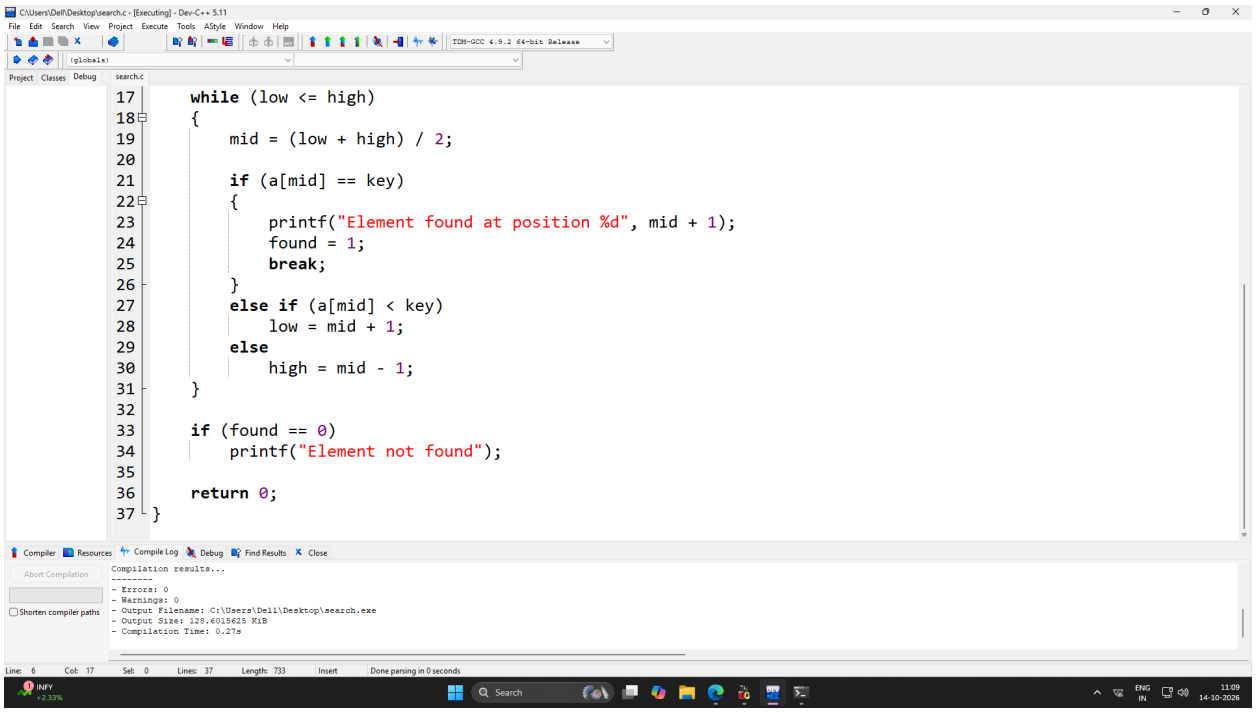
int main()
{
    int a[] = {10, 20, 30, 40, 50};
    int n = 5,i,key;
    int low = 0, high = n - 1, mid;
    int found = 0;

    printf("Array elements: ");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);

    printf("\nEnter element to search: ");
    scanf("%d", &key);

    while (low <= high)
    {
        mid = (low + high) / 2;

        if (a[mid] == key)
        {
```



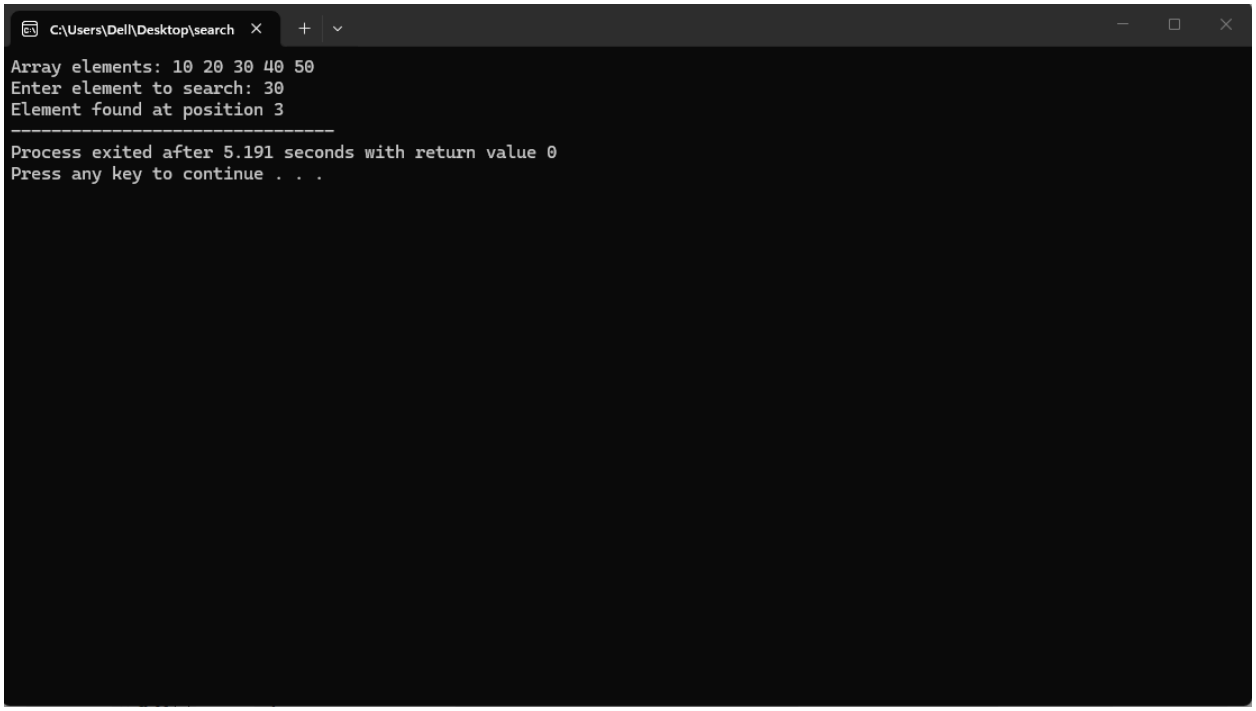
```c
    while (low <= high)
    {
        mid = (low + high) / 2;

        if (a[mid] == key)
        {
            printf("Element found at position %d", mid + 1);
            found = 1;
            break;
        }
        else if (a[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }

    if (found == 0)
        printf("Element not found");

    return 0;
}
```

Output:


C:\Users\Dell\Desktop\search
Array elements: 10 20 30 40 50
Enter element to search: 30
Element found at position 3
--------------------------------
Process exited after 5.191 seconds with return value 0
Press any key to continue . . .